Type 1: Research Project

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Motion Consistency Evaluation in Unmodified Surveillance Video

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1. Problem Statement

Problem Context

Surveillance camera systems generate continuous video streams, but real-world feeds often suffer from hidden issues such as prolonged low motion, frozen frames, or irregular temporal behavior due to environmental or hardware problems. These issues may not trigger traditional object-based alerts yet significantly reduce monitoring reliability.

This project evaluates motion consistency in unaltered surveillance videos by analyzing natural temporal motion patterns to detect irregularities and assess camera feed health and reliability.

2. Literature Survey – Part 1

Related Works (Papers 1–4)

Paper Title	Year	Problem Focus	Essence
Inter-Frame Tampering Detection via Texture & Optical Flow	Akhtar et al., 2024	Inter-frame tampering detection	Proposes a two-stage pipeline combining LBP-based texture analysis with optical flow consistency checks to detect and localize frame-level tampering in surveillance videos.
Blind Video Quality Assessment based on Spatio-Temporal Feature Resolver (STFR-BVQA)	Bi et al., 2024	Blind video quality assessment	Introduces a universal spatio-temporal feature resolver using ZCA and ICA, followed by SVR to predict video quality without relying on deep learning models.
Deep Learning-Based Detection and Localization of Inter-Frame Video Tampering	TBD et al., 2023	Inter-frame inconsistency detection	Employs deep spatio-temporal feature representations to detect and localize frame insertion, deletion, and duplication attacks in surveillance footage.
Frame Duplication Forgery Detection in Surveillance Videos	Li et al., 2023	Frame duplication detection	Uses wavelet-domain analysis combined with texture features to identify and localize duplicated frames in surveillance video sequences.

2. Literature Survey – Part 2

Related Works (Papers 5–9)

Paper Title	Year	Problem Focus	Essence
LSTM-Based Video Quality Prediction for Temporal Distortions	Mittag et al., 2023	Quality prediction with frame freezes/skips.	LSTM models temporal distortions to predict MOS.
Frame Shuffling Forgery Detection Method for MPEG-Coded Video	Kharat J. P., Kharat P. K., 2024	Detection of frame shuffling forgery in MPEG videos.	Uses motion vector discontinuity to detect suspicious frames and validates them using SIFT feature matching.
Vision Transformer Embedded VAD	2025	Deep anomaly detection using transformers	Learns long-range spatial and temporal motion dependencies using transformer-based attention mechanisms.
AnomLite: Efficient Video Anomaly Detection	2025	Lightweight video anomaly detection	Efficient spatio-temporal motion modeling with reduced computational complexity.
Video Quality Assessment with Texture Information Fusion	Menon et al., 2024	Efficient VMAF estimation for streaming.	Uses DCT-based texture features and SSIM fused with an LSTM model to estimate VMAF faster and more efficiently.

3. Research Direction

Existing Approaches

- Surveillance systems primarily focus on detecting **abnormal events or behaviors**.
- Temporal inconsistencies are commonly treated as **scene-level anomalies**.
- Video forensic methods target **intentional frame tampering**.
- Video quality techniques address **network or compression impairments**.
- The **camera feed itself is generally assumed reliable**.

Proposed Research Direction

- Shift from **scene-centric** to **camera-centric** reliability analysis.
- Analyze **natural temporal motion patterns** in surveillance videos.
- Detect silent camera issues such as **freezing, low motion, or instability**.
- No reliance on **object detection or anomaly labels**.
- Applicable to **unmodified real-world surveillance feeds**.

4. Originality - novel idea

- We propose a **camera health assessment method** based on temporal motion pattern analysis in surveillance video streams.
- Unlike traditional approaches focused on object or event detection, our framework enables:
  - Continuous surveillance feed monitoring
  - Non-intrusive evaluation
  - No object detection requirement
  - No human annotation dependency
- The system evaluates **real-world surveillance feeds** purely through motion consistency modeling.

5.Proposed Methodology – Core Concept

Motion Consistency-Based Surveillance Feed Evaluation Framework

- Evaluates **camera health** through temporal motion analysis.
- Focuses on **natural motion consistency**, not activity detection.
- Enables:
  - Continuous monitoring
  - Non-intrusive evaluation
  - No object detection
  - No human annotation

Output: Feed Reliability Score

5. Proposed Methodology – Processing Pipeline

1. **Spatio-temporal feature extraction** from video frames.
2. **Motion representation modeling:**
   - Frame difference
   - Optical flow magnitude
3. **Sliding window temporal analysis**
4. **Deviation detection** for:
   - Frozen frames
   - Frame duplication
   - Motion stagnation
   - Temporal discontinuity

5. Proposed Methodology – Mathematical Model

Motion Estimation

$$M_t = \|F_t - F_{t-1}\| \quad \text{or} \quad M_t = \sum \|\text{Flow}(F_{t-1}, F_t)\|$$

Consistency Measure

$$C_t = \frac{|M_t - \mu_{\text{window}}|}{\sigma_{\text{window}} + \epsilon}$$

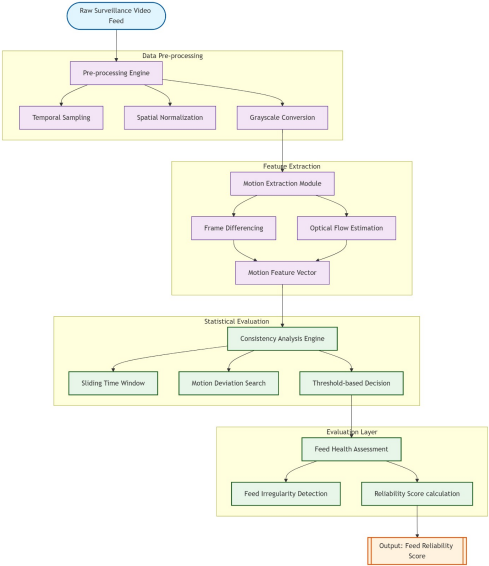
Feed Reliability Score

$$R = 1 - \frac{1}{T} \sum_{t=1}^{T} I(C_t > \tau)$$

6.Limitations of Existing Works

- Designed for **tampering or event detection**, not camera health monitoring.
- Detect specific distortions (duplication, shuffling) rather than overall **motion consistency**.
- Many deep-learning approaches rely on **supervised training**.
- Computationally intensive for **continuous multi-camera deployment**.
- Do not provide a unified **feed reliability score**.

7. Conceptual Architecture- Flow Diagram



8. Dataset Overview

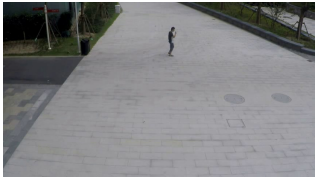
Dataset Used in This Work – ShanghaiTech Campus

Property	Details
Full Name	ShanghaiTech Campus Surveillance Dataset
Type	Real-world outdoor surveillance video
Size	437,271 training frames; 316 test video clips
Scenes	13 distinct camera scenes (fixed, single-scene per camera)
Resolution	Variable per scene; standardized during preprocessing
Frame Rate	24 fps
Anomaly Types	Fighting, chasing, robbery, sudden motion changes
Annotation	Frame-level ground truth for anomalous segments
Key Properties	Static background, real pedestrian motion, diverse lighting
Why Selected	Fixed-camera setup matches our single-scene motion consistency model; natural motion baseline is learnable; contains both normal & degraded motion patterns
Usage in Our Work	Motion baseline modeling, consistency score validation, freeze/stagnation detection benchmarking

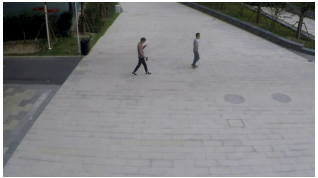
8. Dataset Sample Images

Sample Images from Key Dataset

Frame 0001



Frame 0051



Frame 0103

9. Conclusion & Future Work

Key Contributions

- Introduced a motion-consistency paradigm for surveillance feed reliability assessment.
- Developed a statistical sliding-window model for detecting freeze, duplication, stagnation, and temporal discontinuities.
- Designed a computationally lightweight framework without object detection or human annotation.
- Proposed a unified **Feed Reliability Score** for real-time camera health evaluation.

Future Work (Addressing Limitations)

- Adaptive environment-aware thresholding for dynamic lighting and scene variations.
- Edge-optimized implementation for large-scale multi-camera deployment.
- Incorporating multi-modal features (brightness, noise, compression artifacts).
- Hybrid statistical + lightweight neural temporal modeling.

11. References

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2	Deep Learning for Unsupervised Anomaly Detection *Kiran et al., 2018*	arXiv
3	Blind Video Quality Assessment on Spatio-Temporal Features *Bi et al., 2024*	Neurocomputing
4	Advancing Video Anomaly Detection: Review & New Dataset *Liu et al., 2024*	NeurIPS
5	Multi-camera spatiotemporal deep learning framework for real-time abnormal behavior detection in dense urban environments *S. B. Veesam et al., 2025*	Scientific Reports
6	Lstm-based video quality prediction accounting for temporal distortions in videoconferencing calls, *G. Mittag et al., 2023*	arXiv
7	Frame Duplication Forgery Detection via Textural Features *Li et al., 2023*	Electronics
8	Multi-Camera Spatiotemporal DL for Behavior Detection *Veesam et al., 2025*	Sci. Reports
9	LSTM Video Quality Prediction for Temporal Distortions *Mittag et al., 2023*	arXiv
10	Inter-Frame Tampering Detection via Texture & Optical Flow *Akhtar et al., 2024*	Mathematics
11	Future Frame Prediction for Anomaly Detection *Liu et al., 2018*	–
12	Anomaly Detection via Global-Local Information *Zhang et al., 2023*	ResearchGate

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14	Multi-Sensor Surveillance with Secure Alerting *Hussain et al., 2024*	Sci. Reports
15	SurveillanceVQA-589k: Video-Language Benchmark *Liu et al., 2025*	arXiv
16	Deep-STA: Inter-Frame Tampering Detection *Kaddah et al., 2022*	Electronics
17	Robust Motion Detection Using Motion Vectors *Ali et al., 2023*	PLOS ONE
18	Blind VQA: Spatio-Temporal Resolver *Bi et al., 2024*	Neurocomputing
19	Video QA with Texture Information Fusion *Zhang et al., 2024*	arXiv
20	Frame Shuffling Forgery Detection for MPEG *Wang et al., 2023*	IEEE Access
21	Proactive Anomaly Surveillance for CCTV *Sarker et al., 2024*	Sci. Reports
22	SecureCam: Selective Detection & Encryption *Vora et al., 2023*	IEEE IoT J.

Thank You

Questions & Discussion